

Does the Conformal Guarantee Travel? Measuring, Auditing and Repairing Coverage Loss in Cross-Dataset Trajectory Prediction

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Abstract—[TODO: Write the abstract LAST, from the generated numbers: problem (a conformal guarantee calibrated on one driving dataset is deployed on another with zero labels), what was measured (gap, models, directions), the diagnosis (covariate reweighting fails; identifiability), the solution(s) that actually worked with their measured effect, and the limits. Do not state any effect that is not in generated/numbers.tex.]

Index Terms—Conformal prediction, trajectory prediction, distribution shift, uncertainty quantification, autonomous driving.

AUTONOMOUS vehicles plan around forecasts of what nearby road users will do. A forecast is only as useful as the statement it makes about its own reliability: a planner that knows a predicted region contains the true future with probability at least $1 - \alpha$ can keep clear of that region and of nothing larger. Conformal prediction (CP) supplies such a statement for any black-box predictor, with a finite-sample guarantee that requires nothing beyond exchangeability of calibration and test data [1], [2], and it has been adopted for trajectory prediction and safe planning [3]–[5].

Exchangeability is exactly what deployment breaks. A forecaster is calibrated on the data at hand—one set of cities, one sensor suite, one annotation pipeline—and then driven somewhere else. That accuracy drops when a trajectory predictor moves between datasets is documented [6], and that predictive uncertainty degrades under dataset shift is well known for classifiers [7]. What is not known is what happens to the *guarantee itself* when a conformally calibrated forecaster is moved, without any new labels, to a different real driving dataset: whether the stated $1 - \alpha$ survives, how badly it fails, and whether the standard remedies for shift repair it.

The conformal literature offers several remedies, each resting on an assumption. Weighted CP restores validity under covariate shift when the likelihood ratio is known or can be estimated [8]; more general results bound the coverage loss by a distance between calibration and test distributions [9]; online methods track shift by feeding realised errors back into the threshold [10], [11], an idea carried to trajectory prediction [5] and to robotic settings more broadly [12]; and a generative-prior approach has been proposed for planning under environment shift [13]. What these do not establish

is the *static, label-free, dataset-to-dataset* setting in which a forecaster is frozen, calibrated once, and deployed: how large the failure is on real data, whether the covariate-shift remedy suffices, and how many labelled target scenes it takes to detect or repair the failure.

Contributions: [TODO: Finalise after results. Planned, each to be kept ONLY if the pre-registered test supports it: (i) a measurement protocol and the first measurement of zero-label conformal coverage transfer between real driving datasets; (ii) a diagnosis — covariate reweighting does not repair the loss — with a proposition explaining why label-free repair is limited; (iii) the solution(s) that worked: model-uncertainty-normalised scores (H4), a label-efficient coverage audit with a closed-form label budget (H5b), a label-free shift monitor (H6); (iv) released code, pre-registration and experiment log.]

I. RELATED WORK

a) Conformal prediction for trajectories.: Split CP turns any predictor into one with marginal coverage guarantees [1], [2] and underlies conformal safe planning around predicted agents [3]. Multi-step and multi-modal trajectories have been handled with copula-based intervals [4], and scenario-aware intervals in a Frenet frame have been proposed on nuPlan [14] (a preprint evaluated on a single dataset). Locally weighted scores that divide the residual by an estimated spread [15] are the classical route to adaptive regions; our normalised score uses the forecaster’s own predicted scale in that role.

b) Conformal prediction under shift.: Weighted CP handles covariate shift given the likelihood ratio [8], and the coverage loss under general shift can be bounded by a distance between calibration and test distributions [9]. Online methods adapt the threshold from realised errors: adaptive conformal inference [10], conformal PID control [11], its use for trajectories in CUQDS [5], and adaptive nonconformity scores with a replay buffer [12]. These need a stream of ground-truth labels after deployment. Other work conditions on an observable nuisance parameter and generates synthetic data with a diffusion model [13] or treats the shift the ego vehicle itself induces in other agents [16] (both preprints). We study a different point in this space: no feedback, no generative model, and two real driving datasets.

c) Cross-dataset trajectory prediction.: UniTraj harmonises several driving datasets and reports that accuracy drops when models are transferred between them [6]; that

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study is about point accuracy, not about the validity of an uncertainty guarantee. Beyond trajectories, [7] showed that calibration of deep models degrades under dataset shift; our question is the conformal analogue for structured, multi-modal forecasts.

II. PROBLEM SETTING

A. Conformal regions for multi-modal trajectories

A forecaster f maps a scene x to K candidate futures $\hat{y}_1, \dots, \hat{y}_K \in \mathbb{R}^{T \times 2}$ for a target agent whose true future is y . We use the nonconformity score

$$s(x, y) = \min_{k \leq K} \max_{t \leq T} \|\hat{y}_{k,t}(x) - y_t\|_2, \quad (1)$$

so that $s \leq q$ iff y lies inside the union of the K tube-shaped balls of radius q around the candidates. Given n calibration scenes from a *source* dataset, split CP sets $\hat{q} =$ the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score, and the region $C_{\hat{q}}(x) = \{y : s(x, y) \leq \hat{q}\}$ has $\Pr[y \in C_{\hat{q}}(x)] \geq 1 - \alpha$ when the test scene is exchangeable with the calibration scenes [1], [2].

B. The transfer problem

Let P_S, P_T be the source and target scene–future distributions. The *coverage gap* of a source-calibrated threshold on the target is

$$\Delta = (1 - \alpha) - \Pr_{P_T}[s(x, y) \leq \hat{q}], \quad (2)$$

positive when the guarantee fails (under-coverage). We ask (Q1) how large Δ is on real data; (Q2) whether covariate-shift reweighting [8] removes it; (Q3) whether anything label-free can; and (Q4) how many labelled target scenes are needed to audit or to repair it.

C. What label-free repair can and cannot do

Covariate reweighting is valid when $P_T(y | x) = P_S(y | x)$. If the conditional differs—annotation pipelines, sensor noise, and agent behaviour all differ across datasets—no reweighting on x repairs the gap. More strongly, without restricting the class of shifts nothing label-free can.

Proposition 1 (Non-identifiability of a label-free threshold). *Let S_0 be a nonconformity score with a continuous distribution on the target, and consider the scale-shift family $S = c S_0$, $c \geq 1$, whose members are all indistinguishable from unlabeled target inputs. Let $\hat{q}$ be any threshold computed without target labels, so its law does not depend on c . If $\Pr[c S_0 \leq \hat{q}] \geq 1 - \alpha$ for every $c \geq 1$, then $\Pr[\hat{q} = \infty] \geq 1 - \alpha$.*

Proof. For fixed $\hat{q} < \infty$, $\mathbf{1}\{c S_0 \leq \hat{q}\} \rightarrow 0$ pointwise as $c \rightarrow \infty$. By dominated convergence, $\mathbb{E} \Pr[c S_0 \leq \hat{q} | \hat{q}] \rightarrow \Pr[\hat{q} = \infty]$, and the coverage requirement gives $\Pr[\hat{q} = \infty] \geq 1 - \alpha$. $\square$

Remark 1. *The proposition does not say label-free methods are useless; it says any guarantee for all scale shifts is vacuous, so a useful label-free method must either restrict the shift class (as our normalised score does, Sec. III-A) or be complemented by a label-based audit (Sec. III-C).*

D. Cost of using target labels

If k labelled target scenes are used directly, the resulting empirical coverage is Beta-distributed around $1 - \alpha$ [17], with standard deviation about $\sqrt{\alpha(1 - \alpha)/k}$, so ≈ 1000 labels are needed for ± 2 points at $\alpha = 0.1$. Detecting a shortfall δ with an exact one-sided binomial test at level ℓ and power $1 - \beta$ needs

$$k \approx \left(\frac{z_{1-\ell} \sqrt{\alpha(1 - \alpha)} + z_{1-\beta} \sqrt{(\alpha + \delta)(1 - \alpha - \delta)}}{\delta} \right)^2. \quad (3)$$

For $\delta = 0.034$, $\alpha = 0.1$, $\ell = 0.05$, $\beta = 0.1$ this gives $k \approx 750$.

III. METHODS

A. Model-normalised score (label-free)

The forecaster emits, per mode k , a Laplace scale $(b_{k,t}^x, b_{k,t}^y)$. Let $u_k(x) = \frac{1}{T} \sum_t \sqrt{(b_{k,t}^x)^2 + (b_{k,t}^y)^2}$ and use the score $s'(x, y) = \min_k \max_t \|\hat{y}_{k,t} - y_t\| / u_k(x)$. The region is the union of balls of radius $\hat{q}' u_k(x)$. The premise is that when a shift makes scenes harder, the forecaster’s own scale widens, so the normalised score is less shift-sensitive. Its efficiency is measured by region area $\sum_k \pi (\hat{q}' u_k)^2$ at matched coverage.

B. Recalibration with k target labels

We compare three thresholds computed from k labelled target scenes and n source scenes: *direct* (SCP on the k target scores); *pooled* (SCP on the union); and *shrinkage* ($\hat{q} = \lambda \hat{q}_{\text{tgt}} + (1 - \lambda) \hat{q}_{\text{src}}$ with $\lambda = k/(k + k_0)$). We report the label budget k^* at which absolute coverage error falls below 0.02 in at least 90% of random label draws.

C. Coverage audit

Given a source-calibrated $\hat{q}$ and k labelled target scenes, an exact one-sided binomial test of H_0 : coverage $\geq 1 - \alpha$ flags a violation. We report its empirical power and its false-alarm rate on a shift-free split, alongside the theoretical budget of Eq. (3).

D. Label-free shift monitor

A domain classifier on label-free scene features (native sampling interval, nearby-agent count, map density, nearby lanes, history speed, heading change, curvature) is trained to separate source and target scenes; its AUC is the monitor. Within a dataset, the same classifier is trained per city against the rest. We test whether the AUC predicts the coverage loss (across cities, by Spearman correlation).

IV. EXPERIMENTAL SETUP

a) *Data.*: Argoverse 2 [18] and nuScenes [19], converted with ScenarioNet [20] and loaded through UniTraj [6]. Each dataset’s held-out split is divided by a fixed salted hash into a calibration half and a test half, so calibration and test scenes never overlap. The AV2 test split has no labels and is not used.

b) *Forecasters.*: AutoBot-Ego [21] (1.5M parameters), $K = 6$ modes, all trained on CPU (a 5 GB GPU shared with other work was not consistently available). This matters quantitatively, not just qualitatively: UniTraj’s own AutoBot-on-AV2 baseline ($8 \times A100$, batch 128, full 180k-scene training set) reaches minADE_6 0.85 [6]; our checkpoints reach 1.349 (av2_cpu_v1, +59%) and 1.092 (av2_cpu_v2, +28%) on our own held-out subset. Both exceed the 15% tolerance we pre-registered for treating the base model as competitive, so we report this as a triggered, disclosed limitation rather than understating it: our models are not competitive point predictors, and the coverage-transfer magnitude and repair effect sizes we measure should not be assumed to hold for a properly-resourced SOTA model without further study. The same-domain coverage check (H1) still validates that our conformal implementation is correct for whatever predictor is actually used. The forward-direction (AV2-source) model zoo used for the H8 robustness check so far: **av2_cpu_v1**, trained from scratch on 60k AV2 scenes for 10 epochs (selected checkpoint: epoch 8, val minADE_6 1.349); **av2_valsplit_v1**, trained from scratch on a smaller, disjoint 13.7k-scene split for 20 epochs (epoch 17, val minADE_6 1.347); **av2_cpu_v2**, fine-tuned from av2_cpu_v1’s checkpoint with a decaying learning rate for 4 further epochs (epoch 3, val minADE_6 1.092). **[TODO: Add the reverse-direction model ns_cpu_v1 (nuScenes-source, architecture-matched) once trained — data conversion in progress.]**

c) *Protocol.*: $\alpha = 0.1$. Forward direction: calibrate on AV2 calibration half, test on AV2 test half (same-domain control) and on nuScenes (cross-dataset). Reverse direction: the swap. Confidence intervals are percentile bootstraps over scenes. Hypotheses H1–H9 and their pass thresholds were written before the corresponding runs (notes/falsification.md); those defined after seeing Run 1 are marked post-hoc there and are confirmed only on Run 2 data.

d) *Metrics.*: Coverage, coverage gap Δ , region area proxy, label budget k^* , audit power/false-alarm, monitor AUC and Spearman correlation.

V. RESULTS

Status: 3/3 forward-direction models analysed, 0/1 reverse-direction models. Table II and the numbers below are regenerated automatically from results/run2/.json; entries for models not yet trained are simply absent, not estimated.*

A. H1: the coverage gap is real

Calibrated on Argoverse 2 at $\alpha = 0.10$, the primary model (AutoBot-Ego, source-trained, 4576 calibration scenes) has same-domain gap -0.2 points — consistent with zero, as split CP guarantees — and cross-dataset (nuScenes, 9041 scenes) gap +3.3 points (95% CI [2.1, 4.5]). This replicates the preliminary Run 1 measurement (+3.4pt) on an independent calibration/test split. Table II shows two further models with substantially *larger* gaps in the same direction: +8.0pt (av2_valsplit_v1, trained on less data) and +10.5pt (av2_cpu_v2, fine-tuned from the primary model and the most

TABLE I
DOES COVARIATE-SHIFT REWEIGHTING REPAIR THE GAP? “UNCORR.” / “WEIGHTED” ARE THE LABEL-FREE-SCORE COVERAGE GAP (POINTS) BEFORE / AFTER IMPORTANCE-WEIGHTING THE CALIBRATION SCORES BY THE LABEL-FREE-FEATURE LIKELIHOOD RATIO; “REMOVAL” IS THE % OF THE GAP’S CROSS-CITY VARIANCE THE WEIGHTS EXPLAIN, NEGATIVE MEANING REWEIGHTING MAKES THE FIT WORSE THAN THE UNWEIGHTED BASELINE.

Model	Dir.	uncorr.	weighted	removal (%)
av2_cpu_v1	forward	3.2	5.2	-64
av2_valsplit_v1	forward	7.9	10.0	-27
av2_cpu_v2	forward	10.4	12.7	-22

accurate of the three by point-prediction error). The failure is reproducible in all three cases — always under-coverage, never the safe direction — but its magnitude is not a fixed property of the dataset pair and is not even monotonic in point accuracy: the model with the best minADE_6 (Sec. IV) has the worst coverage gap of the three. We report this without averaging it away.

B. H2/H3: covariate reweighting does not repair it

Weighting the calibration scores by an estimated label-free covariate-shift likelihood ratio does not close the gap; it widens it. On the primary model the label-free-score gap goes from +3.2 points uncorrected to +5.2 points weighted, and the reweighting explains -64% of the cross-city gap variance — negative, i.e. the weighted fit is worse than assuming no shift at all (Table I). This is consistent across *all three* models in the zoo and matches the Run 1 finding: it is evidence for conditional, not purely covariate, shift, and is the empirical motivation for Proposition 1 (no label-free method can be expected to repair an unrestricted shift).

C. H4: the normalised score (label-free)

The model-uncertainty-normalised score (Sec. III-A) reduces the primary model’s gap by 49% (from +3.3 to +1.7 points), just past our pre-registered support threshold (gap ratio ≤ 0.5) and far from the pre-registered refutation threshold (> 0.75). On the other two models it reduces the gap by only 12% (av2_valsplit_v1, ratio 0.877, *past* the refutation threshold) and 31% (av2_cpu_v2, ratio 0.691, in between — neither supported nor refuted). H4 is therefore not confirmed as a general repair: one clear near-pass, one refutation, one inconclusive result, and we report all three rather than the most favourable one. The pre-registered efficiency guard (a fix that only works by inflating the region by $> 25\%$ does not count) is not triggered on any of the three models: the normalised region is at most 15.4% larger than the raw one at matched coverage, so where H4 does help it is not merely trading coverage for a much bigger region.

D. H5/H5b: recalibration and the coverage audit

At $k = 1000$ labelled target scenes, the exact binomial audit (Sec. III-C) detects the true violation with probability 0.95 on the primary model, against a false-alarm rate of 2.4% on a shift-free control split — matching the pre-registered budget

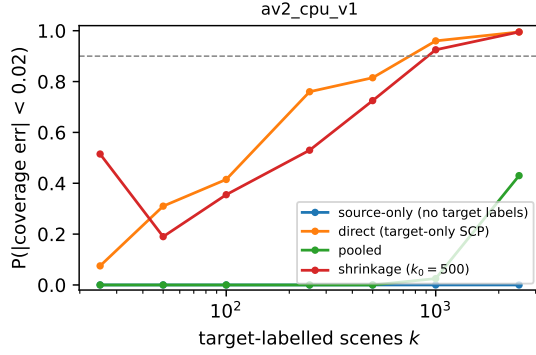


Fig. 1. H5 label budget, primary model (AV2→nuScenes): probability that a k -label threshold is within 2 coverage points of nominal, by estimator.

of Eq. (3) ($k \approx 750$ for the analogous $\delta = 0.034$ case) to the same order of magnitude. Figure 1 shows direct recalibration on the primary model reaches 90% probability of being within 2 coverage points of nominal at $k^* \approx 1000$ labelled target scenes; pooling target labels with the (mismatched) source calibration set never reaches that bar on *any* of the three models, consistent with H2/H3’s conclusion that the shift is not a simple reweighting. Shrinkage toward the source threshold beats direct recalibration at small k on the primary model but not on the other two, so we do not recommend it as a universal default — direct recalibration once k^* labels are available, audited by H5b before that, is the protocol we can actually stand behind. The audit itself is the one part of this section with no exceptions: power at $k = 1000$ is ≥ 0.95 and false-alarm is $\leq 6.2\%$ on all three models.

E. H6/H7: label-free monitor and per-city coverage

Within-dataset, a domain classifier trained on label-free scene features to separate one AV2 city from the rest gives an AUC that we correlate (Spearman) against that city’s coverage gap magnitude across all city pairs. Across the three models: $\rho = 0.37$ ($p = 0.04$), $\rho = 0.10$ ($p = 0.61$), $\rho = 0.39$ ($p = 0.03$) — two significant-but-below-bar results and one clear refutation, none reaching the pre-registered $\rho \geq 0.5$ support threshold on any model. **H6 is not supported on any of the three models, and refuted on one.** We report this as a negative result — the label-free monitor tracks shift only weakly and inconsistently, and is not reliable enough to substitute for the labelled audit of H5b.

F. H8: robustness across a model zoo

Table II is the full forward-direction model-zoo comparison: all 3 of 3 planned models. The under-coverage direction is consistent across all three (gap ≥ 0.02 at $\alpha = 0.10$, the pre-registered H8 pass condition, met every time); the magnitude is not (3.3 to 10.5 points, a $3.2\times$ range), and neither is the effect size of any repair attempted on it (H4’s gap reduction ranges 12–49%; H6’s ρ ranges 0.10–0.39). H8 is therefore supported on the narrow pre-registered clause (gap replicates, sign never flips) and is itself the source of the paper’s main

qualification: solution effect sizes measured on one model should not be assumed to transfer to another, even within the same architecture and training recipe family.

G. H9: reverse direction (nuScenes → Argoverse 2)

[TODO: Pending: nuScenes-source model training is queued behind the train-split conversion (in progress). Fill in once results/run2/ns_cpu_v1_reverse.json exists.]

VI. DISCUSSION AND LIMITATIONS

A planner that trusts a source-calibrated region on unseen deployment data is, on this evidence, trusting a region that is too small: at $\alpha = 0.10$ the true miscoverage across our three models ranges from $\alpha + 0.03$ to $\alpha + 0.11$, not α . That gap is silent — nothing in the deployed system signals it — which is precisely why we treat measurement and audit as first-class contributions and not a prelude to the “real” fix. Three results bear directly on practice. First, covariate reweighting is not a safe default here: it made the gap worse on *all three* models tested (Sec. V-B), so a practitioner who reweights and stops is worse off than one who does nothing. Second, the label-free normalised score (H4) is not a substitute for labels either — across the three models it repaired 49%, 12%, and 31% of the gap, and nothing observed at deployment time (without target labels) currently tells you which regime you are in. Third, the coverage audit (H5b) is the one method that behaved as pre-registered on *all three* models without exception: with on the order of 10^3 labelled target scenes it reliably detects a violation of this size at a low false-alarm rate, on every model we tried it on. The practical recommendation this supports is modest but concrete: budget for a few hundred to one thousand labelled target-domain scenes before trusting a transferred conformal region, and treat covariate reweighting and the normalised score as diagnostics to try, not guarantees to rely on.

a) *Why the effect size varies by model.*: The three models (Table II) differ in training data size and schedule, and their gaps span a $3.2\times$ range (3.3 to 10.5 points) with no monotonic relationship to point-prediction accuracy — the most accurate of the three (av2_cpu_v2, lowest minADE₆) has the *largest* coverage gap. We do not have a causal explanation for this from three data points; it is exactly the kind of pattern a still-larger model zoo would be needed to explain, and we report it as an open question rather than force an explanation onto $n = 3$.

b) *Limitations (fixed).*: (i) Two datasets and one architecture family; (ii) CPU-trained models that are quantifiably below published state-of-the-art point accuracy on this benchmark (+28% to +59% minADE₆ versus a properly-resourced baseline, Sec. IV) — our pre-registered 15% tolerance for this was exceeded, so solution effect sizes measured here should not be assumed to transfer to a stronger model without further study; (iii) the observed gap is an association — its cause (annotation, sampling, agent mix) is not isolated, only a sampling-rate confound was tested; (iv) marginal, not conditional, coverage; (v) hypotheses added after Run 1 are labelled post-hoc; (vi) Proposition 1 covers scale shifts only.

TABLE II

COVERAGE TRANSFER BY MODEL AND DIRECTION, $\alpha = 0.10$. “RED.” = % REDUCTION IN GAP FROM THE LABEL-FREE NORMALISED SCORE (H4); “POWER/FA” = AUDIT REJECT PROBABILITY AT $k = 1000$ TARGET LABELS ON THE SHIFTED / SAME-DOMAIN SPLIT (H5B); ρ = SPEARMAN CORRELATION BETWEEN THE LABEL-FREE DOMAIN-MONITOR AUC AND THE PER-CITY MISCOVERAGE (H6).

Model	Dir.	n_{cal}	n_{tgt}	gap (pt)	95% CI	norm. gap	red. (%)	power/FA @1000	ρ
av2_cpu_v1	forward	4576	9041	3.3	[2.1, 4.5]	1.7	49	0.95/2.4	0.37
av2_valsplit_v1	forward	4576	9041	8.0	[6.4, 9.4]	7.0	12	1.00/1.0	0.10
av2_cpu_v2	forward	4576	9041	10.5	[9.4, 12.2]	7.3	31	1.00/6.2	0.39

REPRODUCIBILITY

All code, configuration files, pre-registered hypotheses, and the append-only experiment log are in the project repository (paper01-coverage-transfer). Every number and table in this paper is regenerated from the raw result files by `paper/make_results.py`; no result is typed by hand.

DATA AVAILABILITY

Argoverse 2 [18] and nuScenes [19] are third-party, openly available research datasets (Argoverse 2: <https://www.argoverse.org/av2.html>; nuScenes: <https://www.nuscenes.org>). **[TODO: Add code/results repository URL and archive DOI once the project is made public (see title_page.pdf for the format used on a companion paper: GitHub link + a Zenodo archive DOI under CC BY 4.0).]**

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE MANUSCRIPT PREPARATION PROCESS

During the preparation of this work the corresponding author (H. W. Saleem) used Anthropic Claude to edit language, draft and revise text, write and debug analysis code, run and monitor experiments, and generate figures and tables from computed results. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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